

```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv(r"C:\Users\Lenovo\Desktop\finalagridata1.csv")
```

```
In [3]: df.head()
```

Out[3]:

	plot_id	location	size	season	crop_type	yield_am
0	108	West Bengal	1490.0	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.0	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.0	Kharif	Soyabean	
3	290	West Bengal	2606.0	Kharif	Arhar/Tur	
4	298	West Bengal	9398.0	Kharif	Ginger	

5 rows × 7 columns

```
In [4]: print(df.dtypes)
```

plot_id	int64
location	object
size	float64
season	object
crop_type	object
yield_amount	float64
fertilizer_used	float64
water_usage	float64
market_price	float64
labour_cost	float64
seed_cost	float64
fertilizer_per_unit	float64
production(quntal)	float64
yield_per_hect(quintal)	float64
total_revenue(hect)	float64
net_profit(hect)	float64
yield_effec(hect,bcr)	float64
fertilizercost_per(hect)	float64
totalcostper(hect)	float64
cooperative roi state/hect	float64
totalrevenue(state)	float64
totalcost(state)	float64
year	int64
STATEYIELD(QUI)	float64
dtype:	object

In []:

In [5]: `df.select_dtypes(include = "object")`

Out[5]:

	location	season	crop_type
0	West Bengal	Kharif	Moong(Green Gram)
1	West Bengal	Kharif	Other Kharif pulses
2	West Bengal	Kharif	Soyabean
3	West Bengal	Kharif	Arhar/Tur
4	West Bengal	Kharif	Ginger
...	...	...	...
995	West Bengal	Kharif	Cotton(lint)
996	West Bengal	Kharif	Jute
997	West Bengal	Kharif	Mesta
998	Karnataka	Kharif	Ginger
999	Karnataka	Kharif	Maize

1000 rows × 3 columns

In [6]:

```
df["market_price"] = pd.to_numeric(df["market_price"], errors="coerce")
df["labour_cost"] = pd.to_numeric(df["labour_cost"], errors="coerce")
df["seed_cost"] = pd.to_numeric(df["seed_cost"], errors="coerce")
df["totalcost(state)"] = pd.to_numeric(df["totalcost(state)"], errors="coerce")
```

In [7]:

```
print(df.dtypes)
```

plot_id	int64
location	object
size	float64
season	object
crop_type	object
yield_amount	float64
fertilizer_used	float64
water_usage	float64
market_price	float64
labour_cost	float64
seed_cost	float64
fertilizer_per_unit	float64
production(qunital)	float64
yield_per_hect(quintal)	float64
total_revenue(hect)	float64
net_profit(hect)	float64
yield_effec(hect,bcr)	float64
fertilizercost_per(hect)	float64
totalcostper(hect)	float64
coperative roi state/hect	float64
totalrevenue(state)	float64
totalcost(state)	float64
year	int64
STATEYIELD(QUI)	float64
dtype:	object

In [8]: `df.head()`

Out[8]:

	plot_id	location	size	season	crop_type	yield_am
0	108	West Bengal	1490.0	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.0	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.0	Kharif	Soyabean	
3	290	West Bengal	2606.0	Kharif	Arhar/Tur	
4	298	West Bengal	9398.0	Kharif	Ginger	

5 rows × 24 columns

```
In [9]: import numpy as np
```

```
In [10]: columns = ["water_usage","fertilizer_per_unit","size","yield"]
for column in columns:
    q1 = df[column].quantile(0.25)
    q3 = df[column].quantile(0.75)
    iqr = q3-q1

    lower=q1-1.5*iqr
    upper = q3+1.5*iqr

    median = df[column].median()

    df[column]=np.where((
        df[column]<lower)|(df[column]>
```

```
upper),median,df[column])  
print("outliers replaced successfully")
```

```
outliers replaced successfully  
outliers replaced successfully  
outliers replaced successfully  
outliers replaced successfully  
outliers replaced successfully  
outliers replaced successfully
```

```
In [11]: df.select_dtypes(include = "number")
```

Out[11]:

	plot_id	size	yield_amount	fertilizer_used	water_u
0	108	1490.0	0.55	141803.0	18
1	113	1279.0	0.59	121722.0	18
2	124	1555.0	1.20	147989.0	18
3	290	2606.0	0.76	257473.0	20
4	298	9398.0	1.45	928522.0	20
...	...	...	...	...	...
995	1483	1429.0	2.55	141442.0	18
996	1492	11951.0	0.81	61409271.0	18
997	1497	9715.0	0.81	961591.0	18
998	1537	11019.0	0.81	1193798.0	18
999	1547	11951.0	3.01	84920684.0	18

1000 rows × 21 columns

In [213]: `df.head()`

Out[213]:

	plot_id	location	size	season	crop_type	yield_am
0	108	West Bengal	1490.0	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.0	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.0	Kharif	Soyabean	
3	290	West Bengal	2606.0	Kharif	Arhar/Tur	
4	298	West Bengal	9398.0	Kharif	Ginger	

5 rows × 24 columns

```
In [224]: x=df[["water_usage","fertilizer_per_unit","size","total_r",  
              "year","fertilizercost_per(hect)","fertilizer_used"  
              y = df["yield_amount"]
```

```
In [225]: x.head()
```



```
Out[225]:
```

	water_usage	fertilizer_per_unit	size	total_revenue(hec
0	1853.0	95.0	1490.0	11753
1	1853.0	95.0	1279.0	24833
2	1853.0	95.0	1555.0	12360
3	2152.0	99.0	2606.0	23803
4	2152.0	99.0	9398.0	19560

```
In [226]: from sklearn.model_selection import train_test_split
```

```
In [227]: x_train,x_test,y_train,y_test = train_test_split(x,y,test
```

```
In [228]: from sklearn.ensemble import RandomForestRegressor
```

```
In [229]: rf = RandomForestRegressor(n_estimators=300,random_state :
rf.fit(x_train,y_train)
```

```
Out[229]:
```

▼ RandomForestRegressor ⓘ ?

► Parameters

```
In [230]: y_pred = rf.predict(x_test)
```

```
In [231]: from sklearn.metrics import mean_absolute_error,r2_score
```

```
In [232]: mae = mean_absolute_error(y_test,y_pred)
print("MAE=", mae)
```

```
MAE= 0.090127166666666684
```

```
In [233]: r2 = r2_score(y_test,y_pred)
print("r2 score =",r2)
```

r2 score = 0.8763272681066478

In []:

In []:

In []: